

TAYLOR PENTZ

Senior Software Engineer

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Professional Summary

Innovative and results-driven Software Engineer with over 9 years of professional experience across full stack development, backend systems, and media pipeline optimization. Proven expertise in PHP, React, and C++ for scalable application design. Adept at debugging complex systems, implementing media ingestion workflows, and architecting resilient solutions using cloud platforms including AWS and GCP. Committed to delivering high-performance code with measurable business value. Passionate about technical excellence, operational reliability, and collaborative development environments.

Technical Skills

- ❖ **Languages & Frameworks:** PHP, C++, React, JavaScript (ES6+), TypeScript, Python, Node.js, HTML5, CSS3
- ❖ **Backend & Architecture:** RESTful APIs, Microservices, Media Stack, Encoding Pipelines, Multithreading
- ❖ **Cloud Platforms:** AWS (EC2, S3, Lambda, CloudWatch), GCP (Compute, Pub/Sub, Cloud Run), Azure
- ❖ **Databases:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB (SQL & NoSQL)
- ❖ **DevOps & Tools:** Docker, Git, Jenkins, Prometheus, Grafana, Webpack, CI/CD pipelines
- ❖ **Other:** Agile, Scrum, Debugging, Observability, Media Ingestion, Encoding Optimization

Professional Experience

SumatoSoft

Senior Software Engineer

Boston, MA (Remote) | Apr 2023 – Mar 2025

- Designed and maintained robust media ingestion services for large-scale streaming platforms, improving encoding pipeline efficiency and reducing latency by 38%.
- Debugged multi-layered system issues involving media codecs, ingestion pipelines, and CDN caching, restoring production stability within SLA-critical windows.
- Led architectural overhaul of a legacy media stack using PHP and C++, delivering 70% improvement in fault tolerance under peak user concurrency.
- Built intelligent alerting and monitoring dashboards using Prometheus and Grafana to detect ingestion anomalies before impacting SLAs or user experience.
- Optimized React-based UIs for media preview and asset handling workflows, enhancing load times by 45% for users across global regions.
- Collaborated cross-functionally with media engineers and QA to simulate complex failure scenarios and harden encoding edge cases.
- Implemented CI/CD pipelines for full-stack deployment, ensuring robust delivery of microservices and frontend modules across environments.

- Mentored junior developers on PHP refactoring strategies and media stream handling, cultivating a culture of code quality and ownership.

Hexaview Technologies Inc.

Software Engineer

New York, NY | Jul 2018 – Mar 2023

- Developed and maintained backend APIs using PHP and C++ for financial and media-based applications, ensuring scalable design across 10+ microservices.
- Troubleshot production issues across backend and frontend layers, resolving database deadlocks, thread contention, and stream decoding faults.
- Engineered a backend queue system for parallel media encoding and ingestion using Redis and MongoDB, cutting processing time by over 50%.
- Integrated media asset workflows with React-based admin interfaces to enable intuitive quality checks and asset tagging for editors.
- Instrumented backend systems with detailed logging and metric tracing, improving incident response time by 62% through real-time anomaly detection.
- Conducted code reviews and performance audits for C++ modules managing file I/O and video parsing to ensure compliance with SLAs.
- Partnered with DevOps to containerize backend services using Docker and improved pipeline automation for QA deployments.
- Led refactoring initiative to separate synchronous and asynchronous ingestion paths, improving resilience against transient upstream failures.

SimbirSoft Company

Software Developer

Boston, MA | Mar 2015 – Jun 2018

- Created full-stack solutions with PHP, React, and JavaScript for enterprise-grade platforms handling user-uploaded media and digital content ingestion.
- Rewrote inefficient modules in C++ for audio/video pre-processing tasks, improving system throughput by 34% under concurrent loads.
- Deployed real-time media ingestion engines with failover handling, enabling uninterrupted processing across high-traffic ingestion nodes.
- Developed integrations for third-party storage providers and CDN caching layers, optimizing media retrieval speeds globally.
- Designed fault-tolerant queues and retry mechanisms for ingestion failures due to malformed or incomplete uploads.
- Built internal dashboards to provide real-time visibility into encoding performance and ingestion status, reducing operational overhead.
- Worked closely with QA to simulate ingestion errors and document edge cases, ensuring production readiness and test coverage.
- Used SQL and NoSQL stores interchangeably for metadata and streaming content logs depending on access patterns and query profiles.

Microsoft

Software Developer Intern

Cambridge, MA | Aug 2014 – Feb 2015

- Assisted in development of internal tools for monitoring video transcoding pipelines and performance dashboards.
- Conducted analysis on encoding job failures and proposed improvements that led to 18% fewer pipeline stalls.
- Built prototype in PHP for a lightweight ingest service simulator to test retry logic and timeout behaviors in video processing chain.
- Collaborated on React dashboard component to visualize media ingestion statuses and encoding queues in real time.
- Benchmarked ingest latency across different media sources, providing valuable insights into network impact and server performance.
- Documented findings and contributed to a team wiki that outlined ingest best practices and error mitigation strategies.
- Participated in code reviews and daily scrums, gaining exposure to professional SDLC and agile methodologies.
- Shadowed senior engineers during incident response calls and debugging sessions to understand real-world operational workflows.

Projects

- ✓ **Real-Time Media Encoding Pipeline (SumatoSoft, 2024)**
Built distributed media encoding engine that scaled ingestion across cloud nodes using PHP and C++, integrated with AWS S3 and GCP Pub/Sub. Enabled parallel processing of 10,000+ media files daily with self-healing and failure retry mechanisms.
- ✓ **Event-Driven Wealth Management System (Hexaview, 2021)**
Led backend design of Kafka-based portfolio alerting system that processed 5M+ events daily. Implemented rules engine for proactive client engagement and risk management.
- ✓ **Full-Stack Media Asset Dashboard (SimbirSoft, 2017)**
Developed end-to-end admin dashboard in React and PHP for asset management, real-time preview, and metadata updates. Enabled internal media teams to verify and validate encoding statuses rapidly.

Certifications

- **AWS Certified Solutions Architect – Associate**
- **Google Associate Cloud Engineer**
- **MongoDB Developer Certification**
- **PHP Zend Certified Engineer (ZCE)**
- **C++ Institute Certified Professional Programmer (CPP)**

Education

University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science (2010 – 2014)